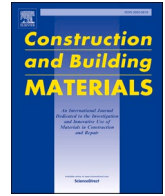




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Evaluation of asphalt pavement texture using multiview stereo reconstruction based on deep learning

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ABSTRACT

To quickly obtain texture information and accurately evaluate the skid resistance of asphalt pavement, a multiview stereo reconstruction method based on deep learning is proposed for evaluating the texture depth of asphalt pavement. First, a depth camera and a digital camera are used to collect RGB-D (i.e., Red, Green, Blue, Depth) and RGB (i.e., Red, Green, Blue) multiview image datasets on different types of pavements for model training and validation respectively. Then, the model precision is validated by calculating the overlap (intersection over union, IoU) between the ground truth point cloud and the reconstructed point cloud. The model performances under different training strategies are compared to obtain the best model, and the effects of image resolution, number of views, and types of pavement material on the model precision are analyzed. Finally, image processing methods and texture depth characterization indexes are proposed to obtain pavement texture information from the reconstructed depth map, thus validating the effectiveness of the model. The results show that the trained model using the transfer learning strategy achieves a reconstruction precision of 0.77 when the image resolution is 3024×3024 and the number of views is 7. Furthermore, the reconstruction performance remains stable across different pavement materials, suggesting the model is suitable for accurately reconstructing depth maps for asphalt pavements. The errors between the predicted texture depth value calculated based on the volume-based index (MTD_v) and the profile-based index (MPD_p) using the depth maps after image processing and the measured texture depth value (MTD_m) using the sand patch method are 11.72% and 18.85%, respectively. It is believed that the pavement texture depth can be effectively evaluated from the reconstructed depth map using the volume-based metric (MTD_v). In contrast to traditional testing methods, this approach requires using only a digital camera and a personal computer, making it a lightweight and intelligent analysis method for obtaining pavement texture depth information.

1. Introduction

The surface texture of asphalt pavement is in direct contact with vehicle tires and crucially impacts the skid resistance function of the pavement [1]. Based on different wavelengths (λ) and amplitudes (A), pavement textures are commonly classified into microtexture, macrotexture, megatexture, or unevenness [2]. Macrotexture on pavement is characterized by wavelengths between 0.5 mm and 50 mm and amplitudes between 0.1 mm and 20 mm; macrotexture is closely related to the skid resistance of the pavement during high-speed vehicle travel [3]. Furthermore, road traffic stands out as the most influential noise source, significantly affecting modern human lifestyles, especially in urban

areas [4–8]. There are numerous factors contributing to its emissions [9], including engines, traffic composition, acoustic impedance, and pavement age [10–13]. Texture, which has only been investigated in recent years alongside mixture properties, plays a pivotal role in this context [14–22]. Consequently, the measurement of macrotexture is crucial for the assessment and prediction of pavement skid resistance and noise emissions.

Traditional methods for measuring pavement macrotexture, such as the sand patch method (SPM) for calculating the mean texture depth (MTD) indicator [23], are subject to human factors and require traffic control during point measurements, which can disrupt normal road usage. These traits limit the accuracy and practicality of the traditional

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methods. Additionally, the mean profile depth (MPD) indicator is typically measured using the laser scanning method [24]. This method requires specialized equipment and expertise, and the measurement process may be affected by environmental conditions and equipment precision limitations. Additionally, the MPD indicator does not reliably predict pavement texture skid resistance [25].

To overcome the limitations of traditional measurement methods, methods related to three-dimensional reconstruction for evaluating pavement texture depth have become a growing interest in recent years. In general, three-dimensional reconstruction methods of pavement texture can be divided into two main categories: active-based and passive-based methods. Active-based methods for pavement texture reconstruction primarily include laser scanning and computed tomography (CT) scanning. Laser scanning refers to the use of a high-precision laser scanner to capture point cloud data of the pavement, which is then employed for the 3D reconstruction of pavement texture and for performance evaluation [26]. This approach offers greater stability and accuracy than traditional MTD measurement methods in various real-world scenarios. However, it requires 3D data as input and is supported by the scanner, which has relatively high usage costs. Furthermore, the complex processing of 3D data reduces the reconstruction efficiency [27]. CT scanning refers to the use of industrial CT scanning technology to obtain three-dimensional digital information on pavement textures. However, this testing method can only scan specimens indoors [28]. Passive-based methods typically involve using high-definition cameras to capture images, which are then reconstructed into pavement textures using digital image processing techniques. Close-range photogrammetry (CRP) technology, for example, captures multiple surrounding images of pavement measurement points from different angles. Through digital feature matching between these images, a three-dimensional model of the pavement texture is reconstructed. The resulting point cloud model of the pavement texture is highly accurate. However, this method requires using a series of post-processing software to reconstruct the 3D model [29]. Multiple photos must be captured by circling the target object to ensure sufficient overlap between the captured images [30], reducing the efficiency of this method. While the close-range photogrammetry method based on binocular vision can improve measurement efficiency, it suffers from fine texture information loss [31,32]. The digital grayscale imaging method involves vertically capturing images of asphalt pavement using a camera. The grayscale values of each pixel in the processed grayscale image can reflect the actual roughness of the pavement. Based on this principle, the grayscale values are converted into texture elevation information, and a three-dimensional pavement model is obtained through 3D reconstruction [33]. However, the computational accuracy of this method is influenced by factors such as lighting conditions, pavement color, and camera position [34].

Due to their relatively simple data structure [27], computation methods based on image processing are popular in practical applications. Deep learning [35] methods for pavement texture reconstruction have advantages in overcoming the limitations of purely visual techniques, which are susceptible to lighting conditions and background noise. In existing studies, pavement textures have been reconstructed from monocular images using a deep convolutional neural network (CNN) or from multiview images with the assistance of the proposed multi-view combination module [36,37]. Furthermore, to address issues such as imbalanced datasets, Generative Adversarial Network (GAN) models are employed for data augmentation and intelligent recognition [38]. These studies demonstrated that deep learning-based reconstruction methods exhibit significant stability and accuracy, despite the influence of pavement materials and other confounding factors. In particular, the reconstruction accuracy of using multiview images is better than that of using monocular images because multiple views provide more texture and illumination information [39], which can compensate for the limitations and uncertainties of a single view. Some other deep learning-based methods require the use of sensors with entry

barriers to obtain texture information. For example, using the AMES profiler (Ames Engineering, Ames, Iowa) to acquire macrotexture data and predict surface friction levels based on CNNs [40], utilizing synthetic aperture radar data and a proposed framework to predict surface roughness [41], obtaining 3D surface texture data using laser scanners and recognizing pavement textures based on the Siamese network [42], or collecting texture data using a 3D Safety Sensor and employing the Pavement Texture Super Resolution Generative Adversarial Network (PT-SRGAN) for texture measurement [43]. These methods may not be as practical in engineering applications as those based on image reconstruction.

The multiview stereo algorithm can infer 3D geometry from a set of images with known positions and perspectives, enabling highly detailed 3D models to be reconstructed based solely on images [44]. This algorithm is typically utilized in depth map, point cloud, mesh, and voxel-based reconstruction methods. Meanwhile, deep learning-based multiview stereo reconstruction methods, which are primarily categorized into depth map-based and voxel-based approaches [45], can overcome the incompleteness of traditional methods. Unlike direct voxel-based reconstruction, which requires high memory consumption [46], depth map-based methods leverage geometric information between views to predict a 2.5D depth map for each view. In this context, a depth map is defined where each pixel value in the image represents the distance from a point in the scene to the camera's xy-plane. The 3D model is then reconstructed through depth map fusion and filtering using 3D fusion techniques. The computational cost depends on only the number and resolution of the input images [45].

The pioneering work in this domain, MVSNet, is an end-to-end deep learning network for depth map inference [47]. It computes only one depth map at a time rather than the entire 3D scene, takes a reference image and several source images as inputs, and infers the depth map of the reference image. PatchmatchNet [48], which is used in this article, is an improvement of the plane-sweeping algorithm [49] used in MVSNet. Since the reference image for depth map prediction is captured parallel to the pavement, the predicted depth is roughly concentrated on a certain depth plane, without excessive foreground or background scenes. Hence, utilizing the plane-sweeping algorithm for estimating pavement texture depth is deemed appropriate. This article further explores this depth map-based multiview stereo reconstruction method for depth texture evaluation.

2. The Objective and roadmap of the work

The objective of this study is to replace the traditional sand patch method with multiview shooting technology and deep learning networks to evaluate the MTD of pavement, shorten the measurement time for pavement texture detection, prevent human error, and achieve accurate and stable predictions for various pavement materials. To achieve fast measurement, PatchmatchNet, a deep learning-based 3D reconstruction model, is used to reconstruct the depth map of the pavement texture. Image processing and texture depth calculation are then achieved without requiring time-consuming dense point cloud reconstruction, providing a new method for evaluating pavement skid resistance.

Specifically, the evaluation of pavement texture depth can be achieved through the following method (red dashed box in Fig. 1): first, an image perpendicular to the pavement is selected from a set of multiview images as the reference view, and the remaining views are used as source views. Then, sparse reconstruction (also known as Structure-from-Motion) is performed to estimate the camera poses of this set of images. Next, multiview images with known camera parameters are used to reconstruct the depth map of the reference view. Finally, the depth value of the pavement texture is obtained through image processing and relevant metric calculations. After multiview images are captured with a camera, relevant information about the pavement texture depth can be obtained solely through computer-assisted calculations.

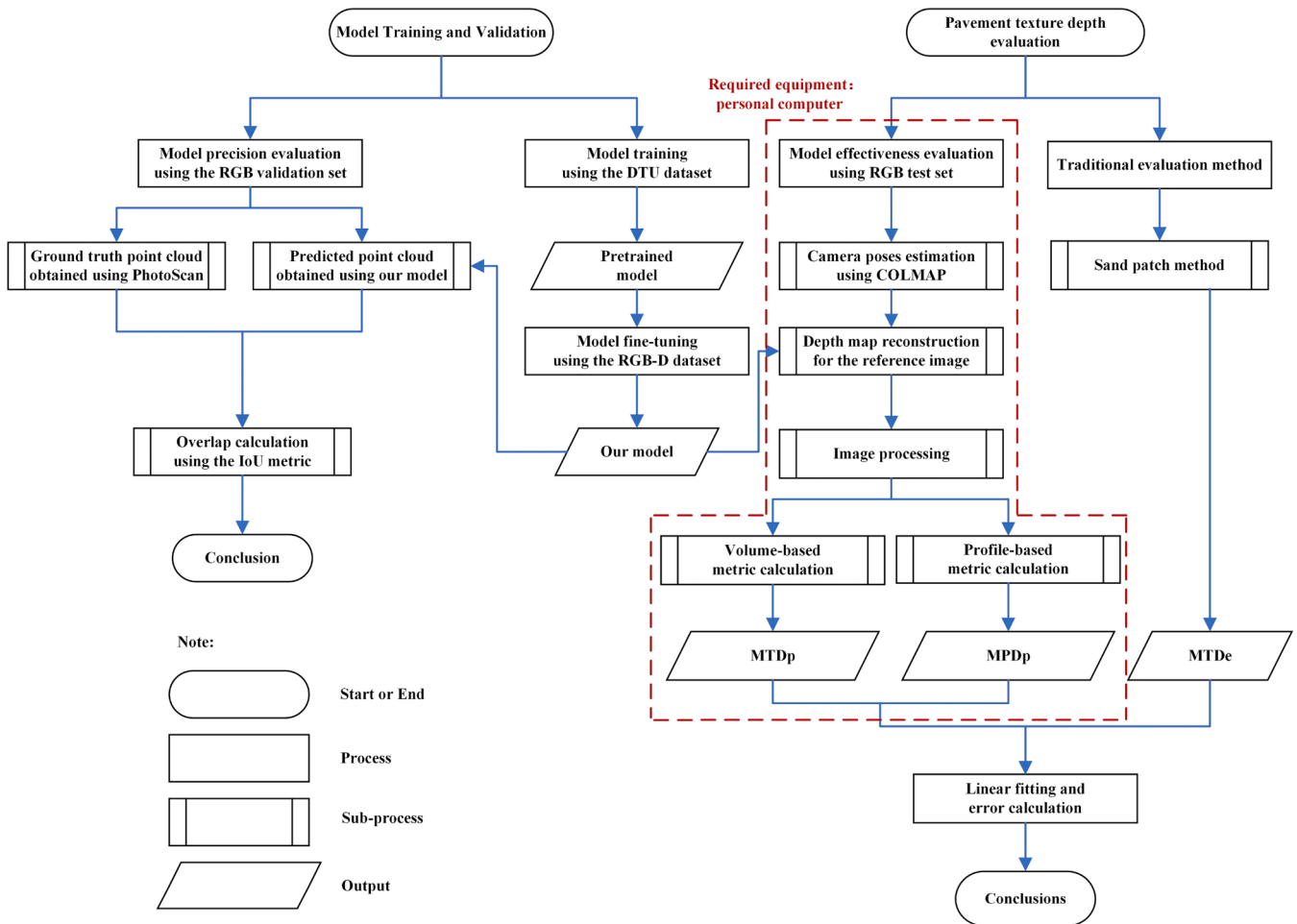


Fig. 1. Roadmap of the research.

The research framework of this article comprises three parts: dataset establishment, model training and validation, and pavement texture depth evaluation. The research routes for the latter two parts are shown in Fig. 1. In Section 3.1, the methods for capturing multiview images using the depth camera and the digital camera are determined, and RGB-D and RGB multiview image datasets are collected on different types of outdoor roads. The dataset is then postprocessed and divided for the subsequent training and validation of the network. In Section 3.2, the precision of the model is verified by calculating the intersection over union (IoU) of the ground truth point cloud and the reconstructed point cloud. The model performances under different training strategies are compared on the RGB validation set to obtain the best model, and the effects of image resolution, number of views, and pavement types on model precision are analyzed. In Section 3.3, a method is proposed to calculate the absolute depth values for the depth map, and an image processing method is proposed to correct tilt errors. Pavement texture depth information is then obtained from the depth map on the RGB test set based on the characterization metrics MTD_p and MPD_p. The model effectiveness is verified by comparison with the measured values of the sand spread method MTD_e.

3. Methodology

3.1. Dataset establishment

The dataset consists of two parts: an RGB-D dataset and an RGB dataset built from scratch. To facilitate fixed-point shooting and the subsequent recovery of absolute depth values from the depth map, a

calibration board measuring 100 mm × 100 mm with a thickness of 3 mm was used (Fig. 2(c)). A total of 10 different asphalt pavements with varying service times in Chongqing and Changsha were selected. Among these, six were chosen to create the RGB-D dataset, while the remaining four were used to create the RGB dataset. Some example road landscapes are shown in Fig. 3(a). Fig. 4 illustrates the shooting viewpoints of the reference image and some partial source images, with the requirement that the overlap area between adjacent frames should be no less than 70% during shooting. This criterion is crucial for multiview depth estimation, as it relies on the overlapping areas between multiple images. To ensure that the dataset covers most common asphalt pavement materials, four common pavement types were selected, including asphalt concrete pavement (AC-13 and AC-16), stone mastic asphalt pavement (SMA-13), and open-graded friction course pavement (OGFC-16). The dataset was captured during daylight with sufficient light to avoid external shadows. In Table 1, the characteristics of the pavements from which data were collected for the study are presented. This table provides information on pavement types, ages, and the number of RGB and RGB-D images captured from each type of pavement. These details are essential for gaining insight into the diversity of the dataset and its relevance to the research objectives.

3.1.1. RGB-D dataset

Since the performance of deep learning networks largely depends on the quantity and quality of training samples, having as much high-quality training data as possible is essential. However, large-scale public datasets such as the DTU [50], BlendedMVS [51], and ScanNet [52] datasets include captures of indoor and outdoor macro objects, which do

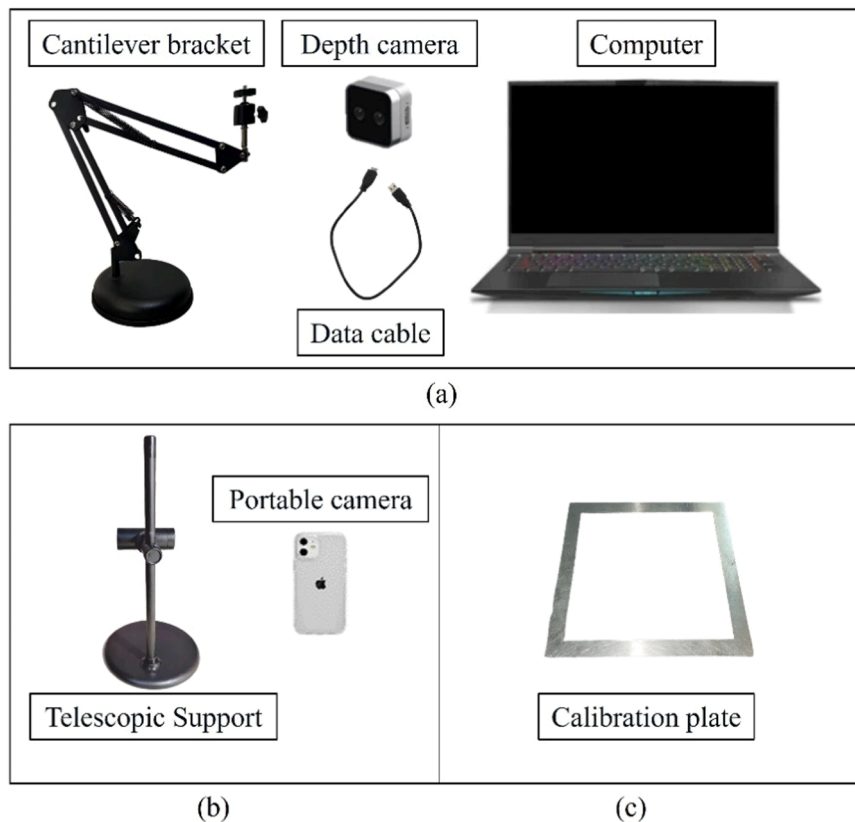


Fig. 2. (a) RGB-D dataset acquisition facilities; (b) RGB data capture equipments; (c) Calibration plate.

not match the scale of the pavement texture depth to be evaluated. To improve the accuracy and scene specificity of the model's predictions, an RGB-D dataset was collected for asphalt pavements. The equipment used to capture the RGB-D dataset is shown in Fig. 2(a). The Intel RealSense D405 is a depth camera based on Intel's depth sensing technology that simultaneously captures three-dimensional depth data and RGB data using active infrared stereo technology, thus providing a pair of RGB-D training samples. The camera provides up to 1280×720 pixel resolution and a 60 frames per second frame rate; it can achieve accuracy within 1 mm in close-range shooting, which essentially meets the requirements of pavement texture depth calculation. The camera was mounted on a compatible bracket and controlled by a computer program to capture images from multiple viewpoints at a distance of approximately 15 cm from the ground, while avoiding image blur caused by unnecessary shaking. To ensure the accuracy of the depth map information, overexposure and underexposure were avoided. Approximately 30 pairs of RGB-D images with a fixed resolution of 640×640 were captured for each calibration board. Postprocessing techniques such as hole filling and depth value smoothing [53] were used during shooting, and alignment scripts were used to register image pairs, resulting in an RGB-D image set with the same resolution and a one-to-one correspondence between pixel points.

The depth map captured by a consumer-grade RGB-D camera can provide depth information to some extent, but its measurements exhibit errors and uncertainties, leading to the loss of some depth details. Additionally, the depth capture range of the sensor is limited to a certain distance, which hinders its use in close-range scenarios. Furthermore, due to the influence of ambient light during the collection process, the obtained pavement texture depth map obtained is poor quality, so postprocessing is necessary before the map is input into the network for training. Most depth completion algorithms based on deep learning are designed for transparent and reflective objects, so they are not suitable for this dataset. Therefore, the IP-Basic algorithm based on traditional

computer vision methods was used to complete the depth map by filling holes and removing noise [54], making the depth map complete and continuous and providing a relatively accurate initial depth value for training. In subsequent research, it was found that models trained on a higher resolution RGB-D dataset can more accurately reconstruct the depth map. Therefore, the RGB-D dataset was upsampled to a resolution of 1000×1000 using the bilinear interpolation algorithm [55]. Then, the Open3D library was used for pose estimation of the RGB-D dataset, and 2200 pairs of RGB-D images were obtained for subsequent model training. Examples of RGB-D image pairs are shown in Fig. 3(b).

3.1.2. RGB dataset

The RGB-D dataset is not used to evaluate the model performance because the depth map captured by the depth camera may have missing depth values. Even after depth completion, this depth map cannot be used as the ground truth for validation. In multiview stereo reconstruction based on depth maps, the quality of the depth map directly affects the accuracy of the reconstructed point cloud. Therefore, the accuracy of the depth map reconstruction can be indirectly evaluated by analyzing the similarity between the reconstructed point cloud and the real scene. Previous studies [30] have shown that traditional multiview stereo reconstruction techniques based on photogrammetry exhibit satisfactory accuracy, and the predicted pavement texture depth strongly correlates with the measured values. Therefore, the ground truth for validation in this study is taken from the point cloud accurately reconstructed from the RGB dataset using specialized photogrammetry software, PhotoScan [56], rather than the depth map, and the validation object is the point cloud obtained by fusing the predicted depth maps from multiple views of a scene. A strong precision validation indicates that the model can reconstruct high-quality depth maps. Additionally, the RGB dataset is also used for the effectiveness evaluation of the model, so building a well-constructed RGB dataset is crucial.

To meet the development needs of subsequent mobile testing

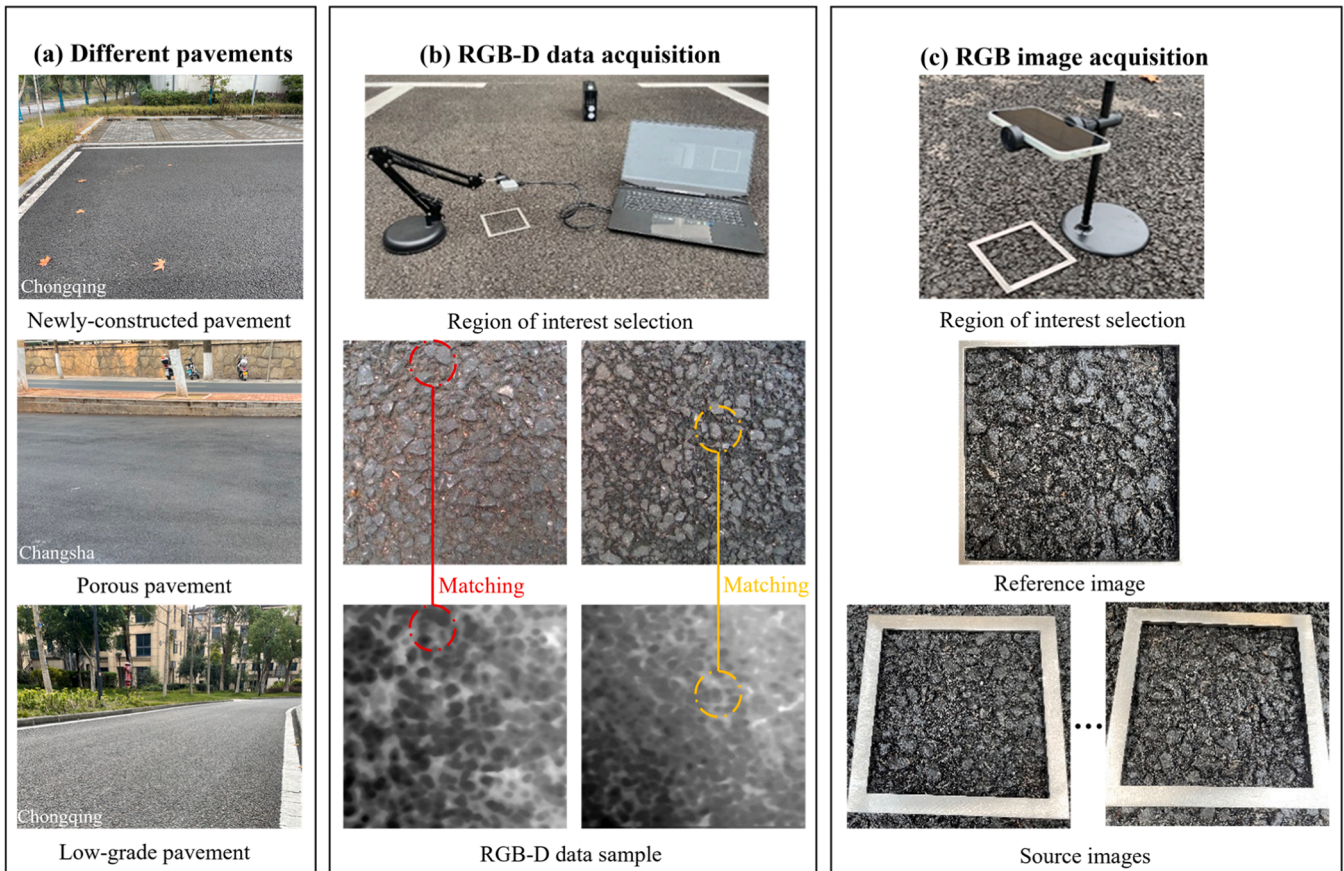


Fig. 3. The process of data collection.

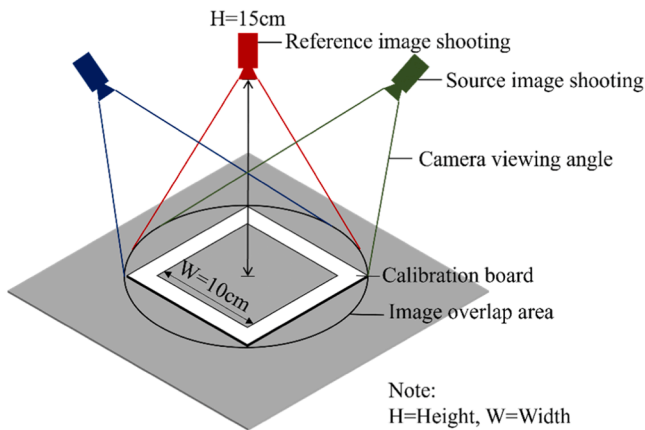


Fig. 4. Camera poses during image acquisition.

devices, an iPhone 12 was used to capture the RGB dataset, and a phone holder was used to assist with the shooting (Fig. 2(b)). Similarly, images were captured from multiple viewpoints at a distance of approximately 15 cm from the ground, and Bluetooth remote control was used for capturing the images to avoid motion blur. When shooting the pavement on the calibration board, the camera's viewpoint in the reference image should be parallel to the pavement, and the camera's viewpoint in the source image should be evenly distributed around the reference image to provide sufficient multiview texture information (Fig. 3(c)). The number of images in each group of data should not be overly large, as this would result in a long data collection time not suitable for efficient engineering applications; this number should also not be overly small, as this would

Table 1

Pavement data characteristics.

RGB-D dataset			RGB dataset		
Type	Age	Number of image pairs	Type	Age	Number of image sets
AC-13	5 years	260	AC-13	4 years	600
AC-13	2 years	290	AC-16	3 years	600
AC-16	7 years	230	SMA-13	5 years	600
AC-16	2 years	310	OGFC-16	2 years	600
SMA-13	4 years	550			
OGFC-16	1 year	560			
Total		2200			2400

affect the precision of the depth map reconstruction. The camera's parameters, such as aperture and shutter speed, should be controlled to ensure consistent exposure between different images. A total of 240 sets of RGB datasets of different types of roads were captured, with a fixed number of 10 images per group and a resolution of 3024×3024 for each image.

The RGB dataset was divided into validation and test sets in a 1:1 ratio. These sets were used to evaluate the model's precision and effectiveness, respectively. To evaluate the model's precision at different image resolutions, the RGB validation set was downsampled twice and upsampled once while the original resolution of 3024×3024 was maintained, resulting in resolutions of 2048×2048 , 2560×2560 , and 3400×3400 .

3.2. Model training and validation

3.2.1. Precision evaluation metric

To evaluate the precision of the model on the RGB validation set, the IoU [57] evaluation metric was introduced to quantitatively analyze the overlap between the predicted point cloud and the ground truth point cloud. The IoU is defined by Eq. (1):

$$IoU = \frac{V_{GT \cap Pre}}{V_{GT \cup Pre}} \quad (1)$$

where *GT* refers to the point cloud reconstructed using PhotoScan software, which is considered as the ground truth. *Pre* refers to the point cloud reconstructed using the model, which is used as the evaluation object for precision. Calculating the IoU involves converting the point cloud into a three-dimensional grid representation with voxels as the basic unit. The voxelization process assigns the points in the point cloud to corresponding voxels and calculates the ratio of the intersection of the two point cloud voxels to their union. IoU values range from 0 to 1. The closer IoU value is to 1, the more similar the spatial shapes of the two 3D models are, indicating that the precision of the predicted point cloud is better and that the model can perform good quality depth map reconstruction. The specific process can be seen in Fig. 5. In the first stage, the reference image along with its N-1 source images form the inputs of the network and the reconstructed depth maps are obtained by using our model. Then, all the depth maps are fused to create a point cloud. In the second stage, the same set of images is input into the PhotoScan software and reconstructed to a point cloud, which is used as the ground truth point cloud. In the third stage, the IoU metric is used to calculate the overlap area between two point clouds.

3.2.2. Training strategies

3.2.2.1. Training on the established pavement texture RGB-D dataset directly. The model was trained directly on the RGB-D dataset on an NVIDIA GeForce RTX 3080 laptop device with 16.0 GB of GPU memory. To obtain the best-performing model, multiple experiments with

different hyperparameter settings were conducted. The training parameters resulting in the best model performance are shown in Table 2. During training, the data loading module randomly scaled and cropped the input images. To enhance the robustness of the model, a training strategy was adopted that randomly selects images from the dataset to match the reference image, rather than selecting the source image that best matches the reference image. In addition, segmented learning rate decay was used to stabilize the model as it approaches the optimal solution. The initial learning rate was set to 0.0001, and the learning rate was reduced by half after the 10th, 12th, and 14th training epochs. Weight decay regularization was also used to reduce the overfitting risk, with the parameter set to 0.0005 to penalize larger weight values and avoid overly fit and constrained models. The Adam optimization algorithm with an adaptive learning rate was used to accelerate the training process, enabling the model to converge more quickly to a local optimal solution and avoiding unstable training, gradient vanishing, or explosion caused by a learning rate that is overly large or small for gradient descent algorithms.

3.2.2.2. Training by the transfer learning method. When obtaining the training dataset for a neural network is difficult, transfer learning [58] can be used for model training. Transfer learning applies knowledge acquired from one task to other related tasks to improve performance on the target task, especially when data for the target task is limited or hard to obtain. It helps models adapt to new tasks more quickly, reducing training time and resource costs. Due to time and labor limitations, the amount of RGB-D data collected in this study is smaller than that in

Table 2

Parameter settings for training on the RGB-D dataset.

Parameter	Value	Parameter	Value
image_max_dim	1000 × 1000	n_views	7
batch_size	1	epoch	16
learning rate	0.0001	lreps	10,12,14:2
weight decay	0.0005	robust_train	True

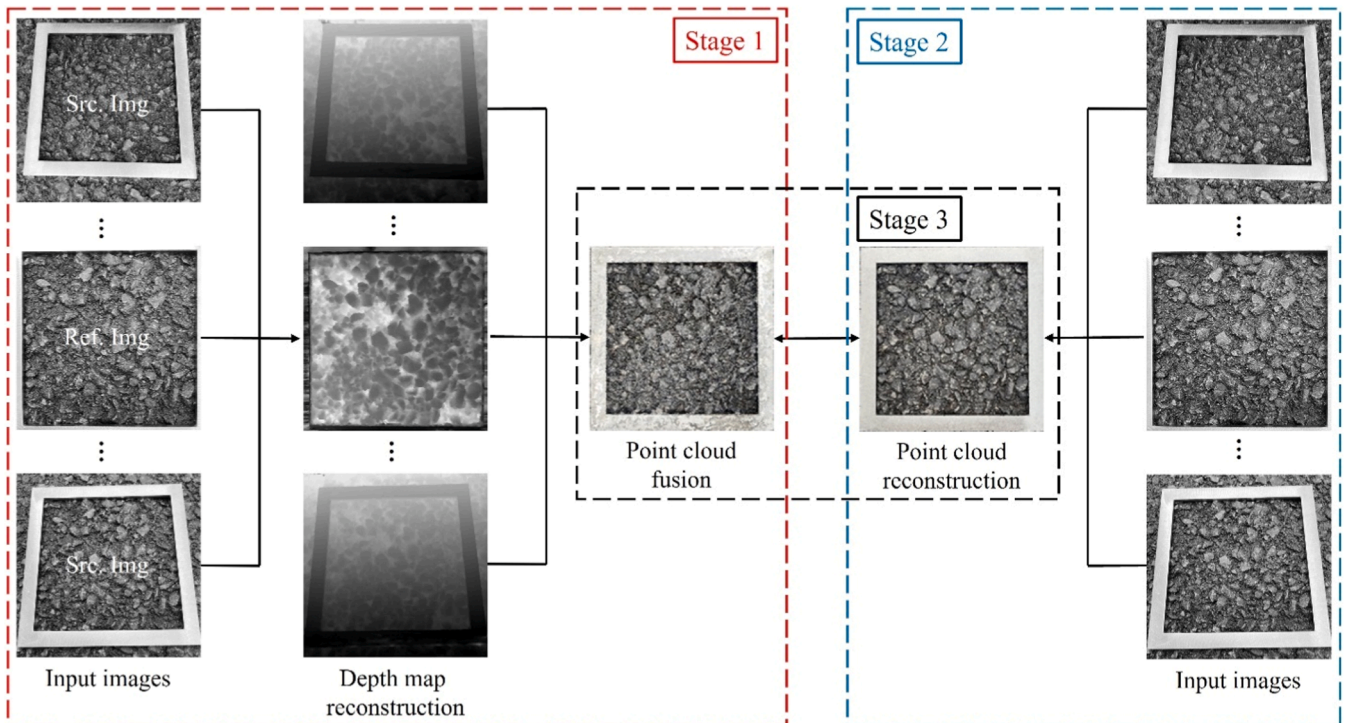


Fig. 5. The process of model precision validation.

publicly available datasets. To overcome this problem, we consider using the large-scale public dataset DTU for pretraining the model, using the pretrained model's weights as the initialization parameters, and then fine-tuning the model on the RGB-D dataset [59]. This approach can use the features and parameter initialization values learned by the pretrained model to accelerate model training and improve performance while reducing the data requirements and training time. The model further learns the pavement texture features from the RGB-D dataset based on the pretrained weights and is expected to achieve good reconstruction results.

The DTU dataset [50] contains 128 controlled laboratory environments, covering various objects and materials. Each scene was scanned using a structured light scanner at 49 camera positions under 7 different lighting conditions, resulting in a total of 43904 pairs of RGB-D data with a resolution of 1200×1600 pixels. When the DTU dataset was used for pretraining, the maximum image size of the input image was set to 640×512 in consideration of computational resources and model training efficiency. Four pairs of data were sent to the neural network for training at each iteration. Because of the large amount of data, the learning rate was set to 0.001 to accelerate the model's convergence, and segmented learning rate decay was also used. The weight decay value was set to 0.005 to limit the model's complexity and improve its generalizability. The specific training parameters are shown in Table 3. The model was then fine-tuned using the pavement texture RGB-D dataset with the same process as described in Section 3.2.2.1.

3.2.3. Analysis of factors affecting model precision

The completeness and accuracy of the reconstructed point clouds both affect the degree of overlap (IoU) with the ground truth point cloud. Accuracy refers to the degree to which the reconstructed model matches the texture features of the real object. Highly accurate 3D reconstruction requires high-resolution, distortion-free input images. Completeness refers all objects and details in the scene being included in the reconstructed result, without omissions or information loss. High completeness in 3D reconstruction requires comprehensive data and complete viewing angle information. However, due to hardware limitations, a trade-off between accuracy and completeness often exists. Pursuing higher accuracy may require more views to provide texture information; otherwise, the reconstruction completeness will be impacted. Similarly, pursuing higher completeness limits the model's ability to process high-resolution images, thereby affecting the accuracy of reconstruction. Therefore, evaluating the impact of different numbers of views and image resolutions on the model reconstruction quality is necessary.

To further evaluate the reconstruction precision of the model on different pavement materials, the data in the RGB validation set, given an image resolution of 3024×3024 , were classified according to pavement materials, and the set was reconstructed with a fixed number of 7 viewpoints. Both quantitative analysis of the point cloud reconstruction results and qualitative analysis of the depth map reconstruction results were conducted.

3.2.4. Comparative studies of different models

A comparison study was conducted to demonstrate the advantages of the proposed framework on the 3D reconstruction of pavement texture. Four additional models were selected in the study. The first model, MVSNet [47], represents the initial generation of learning-based 3D

reconstruction models. The second model, JDACS-MS [60], is an unsupervised depth estimation model that provides reliable guidance for depth estimation through additional priors of semantic consistency and data augmentation consistency. The third model, Fast-MVSNet [61], improves the reconstruction speed by using a prediction framework from sparse to dense. The fourth model, COLMAP [62,63], is a traditional 3D reconstruction method that restores point clouds through sparse reconstruction, depth map estimation, and dense reconstruction. For fairness, the same batch of RGB validation sets was used for comparison, and the same image resolution and number of viewpoints were used in each model.

3.3. Pavement texture depth evaluation

To validate the effectiveness of the model, a method for analyzing and comparing the depth values of reconstructed textures and on-site measured values through image processing and metric calculation is proposed in this section. Evaluation was conducted on the RGB test set containing different pavement materials, with a fixed multiview image resolution of 3024×3024 for the input model and a fixed number of 7 views. Notably, the RGB test set used in this section was collected from different pavements than the RGB validation set discussed earlier.

3.3.1. Image processing

Fig. 6 shows the specific process of the image processing stage, which aims to obtain a depth map that can be directly used for metric calculation. Each step is discussed in detail in this section.

3.3.1.1. Scale restoration and ROI selection. Generally, to uniquely describe the coordinates of each spatial point and the position of the camera, a world coordinate system must first be defined. Then, to establish the mapping relationship between 3D space points and the camera plane, as well as the relative relationship between multiple cameras, a local camera coordinate system is defined for each camera. To establish geometric constraints, the structure-from-motion (SFM) technique [62] is used to recover the camera poses from multiple views, so that the MVS algorithm can infer the depth information of the scene based on the geometric relationship between images. In this paper, COLMAP open-source software is used as a mature SFM technique, which can accurately estimate the camera's intrinsic and extrinsic parameters. COLMAP defaults to using the camera coordinate system of the first image in the multiview image set as the world coordinate system, with the camera facing the negative Z-axis. Points closer to the camera have smaller depth values (Fig. 7). Therefore, the reference image should be captured first, and the optical axis of the reference view should be as perpendicular to the pavement as possible to make the depth values in the depth map correspond to the depth of the pavement. However, the reconstructed depth map cannot be directly used to calculate the depth values of the pavement because without calibration information, the depth values in the reconstructed depth map are defined as the Z values of the camera coordinate system rather than the absolute depth values in the real world. Therefore, to obtain absolute depth values, additional information and constraints are required for scale calibration. Four pairs of calibration points (marked with blue and red dots) are taken at the upper and lower sides of the four corners on the calibration board with a known thickness of 3 mm (Fig. 8(b)). The scale factor can be calculated, and the absolute depth values can be obtained using Eq. (2):

$$Z_{abs} = Z_{rel} \cdot scale = Z_{rel} \cdot \frac{1}{4} \sum_{i=1}^4 \frac{3mm}{Z_{blue,i} - Z_{red,i}} \quad (2)$$

where Z_{abs} represents the absolute depth value, Z_{rel} represents the relative depth value, $scale$ is the scale factor, and $Z_{blue,i}$ and $Z_{red,i}$ are the relative depth values of each pair of calibration points. For ease of writing, the term "Z" is used instead of " Z_{abs} ", and the term "depth

Table 3

Parameter settings for training on the DTU dataset.

Parameter	Value	Parameter	Value
image_max_dim	640×512	n_views	5
batch_size	4	epoch	16
learning rate	0.001	lreprochs	10,12,14:2
weight decay	0.005	robust_train	True

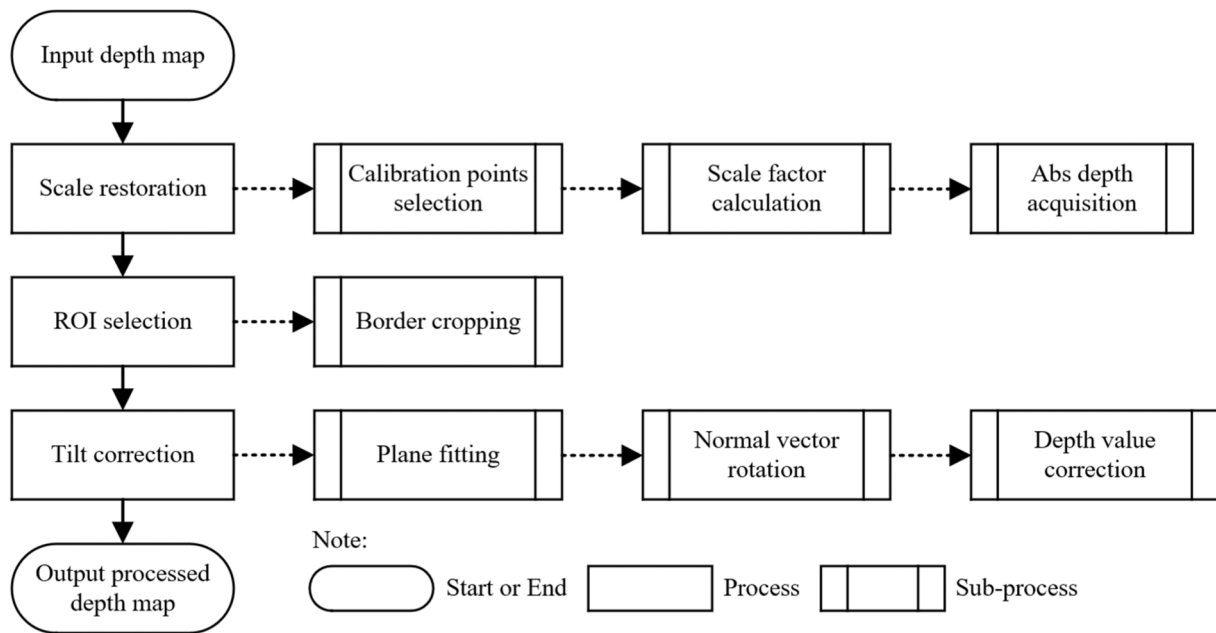


Fig. 6. The workflow of image processing.

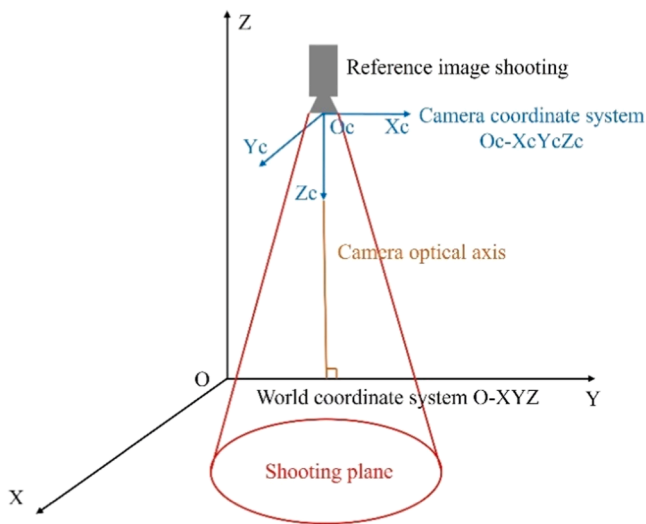


Fig. 7. Coordinate system definition.

value" is used instead of "absolute depth value" in the following text.

After the scale-recovered depth map is obtained, the image must be cropped according to the region of interest (Fig. 8(c)) defined by the calibration board (Fig. 8(d)). Cropping the image as close as possible to the 100 mm × 100 mm border inside the calibration board is important for ensuring the real size of each pixel in the depth map is reliably and accurately calculated.

3.3.1.2. Tilt error correction. Due to the possibility of the camera being tilted or in an abnormal position during shooting, and the presence of road slopes, the optical axis of the reference image cannot be guaranteed to be completely perpendicular to the pavement. Therefore, the inclination of the pavement normal vector must be corrected, and the pavement normal vector must be rotated to align with the Z-axis of the camera coordinate system. Specifically, the RANSAC algorithm [64] is first used to fit a plane to the depth map and obtain the normal vector n of the fitted plane. Then, a coordinate transformation matrix T is constructed to project the points in the depth map to the rotated coordinate system, thus obtaining the depth map with the corrected pavement normal vector (Eq. (3)):

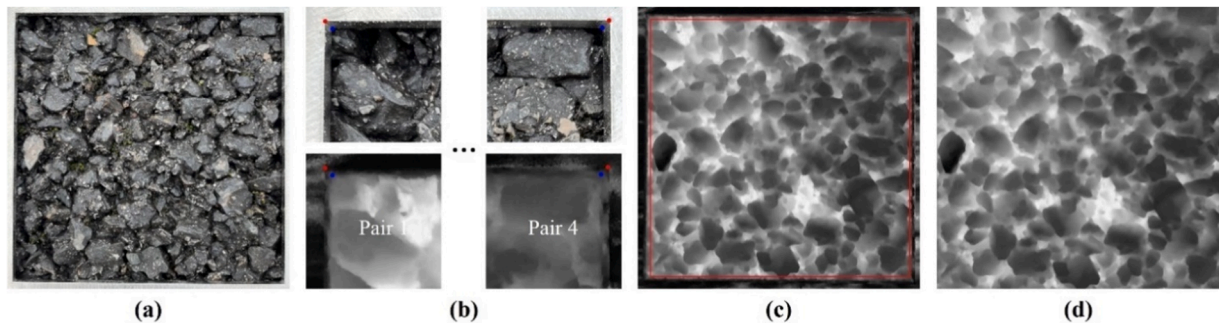


Fig. 8. Depth map scale recovery and cropping: (a) Reference image; (b) Selection of 4 pairs of calibration points marked with red and blue colors; (c) ROI selection; (d) Depth map after border cropping.

$$Z' = \left(T \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \right)_2 = \left(\begin{pmatrix} R & t \\ 0_3^T & 1 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \right)_2 \quad (3)$$

where Z' represents the corrected depth value and X, Y, Z are the pixel coordinates and the corresponding depth value in the original depth map, respectively. If the rotation axis $r = n \times \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$ is a zero vector, then the rotation matrix R is the identity matrix. Otherwise, $R = \text{Rodrigues}\left(\frac{r}{\|r\|} \cdot \theta\right) [0]$, where $\theta = \arccos\left(n \cdot \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} / \|n\|\right)$ is the angle between n and the Z -axis and t is the three-dimensional translation vector.

Fig. 9(a) shows the depth value changes of a certain profile before and after the correction of tilt error, and Fig. 9(b and c) show the corresponding depth maps. In the same depth map, the darker the color of a region is, the smaller the depth value of the points in that region, which indicates that the texture is protruding from the surface. The lighter the color of a region is, the larger the depth value of the points in that region, which indicates the texture is sinking into the surface. The color difference of the depth map after the correction of tilt error is smaller than that of the depth map before correction, indicating that the absolute difference of the depth value is reduced. The overall slope of the depth value profile is also reduced, indicating that the tilt error of the pavement has been effectively corrected.

3.3.2. Effectiveness evaluation metrics

To determine the effectiveness of the model, whether the reconstructed texture features are consistent with the corresponding actual pavement texture must be determined. If a significant correlation exhibits between the two, the reconstructed texture can be considered effective [65].

3.3.2.1. Benchmark metric. MTD is a three-dimensional analysis index that considers the spatial characteristics of pavement texture. It calculates the mean texture depth of the pavement by gradually spreading fine particles with a volume of V onto the pavement, forming a circle with a diameter of D , as shown in Eq. (4). To measure MTD within the

calibration board range, an improved sand patch method (Fig. 10) is used to measure the texture depth in the field [36]. Testing is preferably conducted on dry and clean areas of the pavement, and different test points are selected for each road. 25 cm^3 of standard sand, measured by a measuring cylinder, is gradually poured into the target fixed area. Simultaneously, standard sand is spread in the sample area using a tool, and pouring is stopped when the standard sand fills the calibration board square. The final reading v' of the remaining standard sand in the measuring cylinder is recorded using a graduated cylinder, and the measured mean texture depth MTD_e is obtained using Eq. (5).

$$\text{MTD} = \frac{4V}{\pi D^2} \quad (4)$$

$$\text{MTD}_e = \frac{(25 - v') \cdot 10^3}{100^2} \quad (5)$$

3.3.2.2. Volume-based metric. To accurately characterize the depth values of the reconstructed texture, appropriate calculation indexes must be selected. The depth information of each pixel in the depth map is the depth value relative to the imaging plane of the camera, but what needs to be calculated is the depth value of each pixel relative to the texture reference plane [30]. To avoid the influence of extremely small depth values (elevation peaks) on the calculation, the plane containing the p % smallest depth percentile value in the depth map is selected as the texture reference plane (Fig. 11). The mean depth value MTD_p relative to the texture reference plane is then calculated for the remaining $100-p$ % pixels using Eq. (6) to reflect the overall trend of the pavement texture variation.

$$\text{MTD}_p = \frac{\sum_{m=1}^M \sum_{n=1}^N (Z_{mn} - Z_p) \cdot s}{S} \quad (6)$$

$$s = \frac{100^2}{M \cdot N} \text{ mm}^2$$

where M and N are the number of pixels in the depth map in the longitudinal and transverse directions, respectively; Z_{mn} represents the depth value of the pixel in the m -th row and n -th column; Z_p represents the depth value of the selected texture reference plane; $S = 100^2 \text{ mm}^2$ represents the area of the calibration board; and s represents the size of each pixel in the depth map.

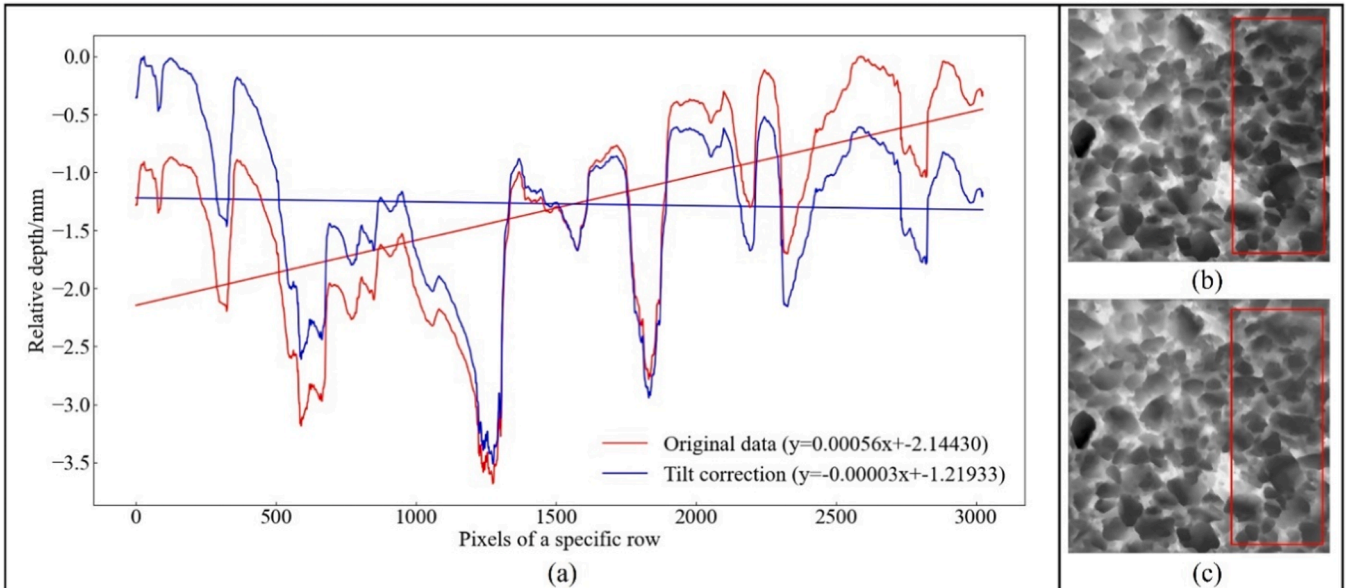


Fig. 9. (a) Profile depth values of the 1060th row; (b) Original depth map; (c) Depth map after tilt error correction.

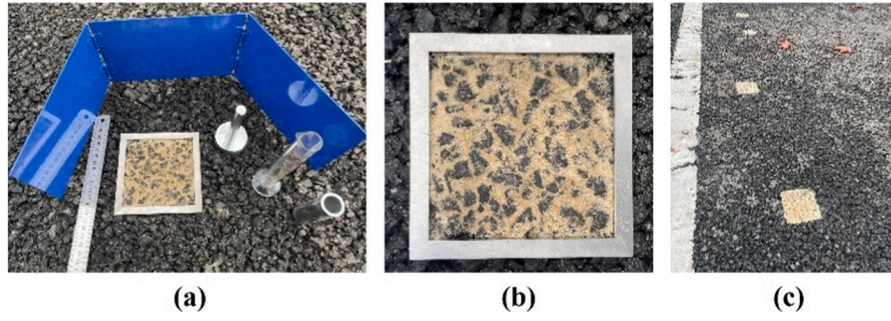


Fig. 10. Measurement of MTD_e using sand patch method: (a) Experimental equipments; (b) Measuring range; (c) Fixed-point measurement.

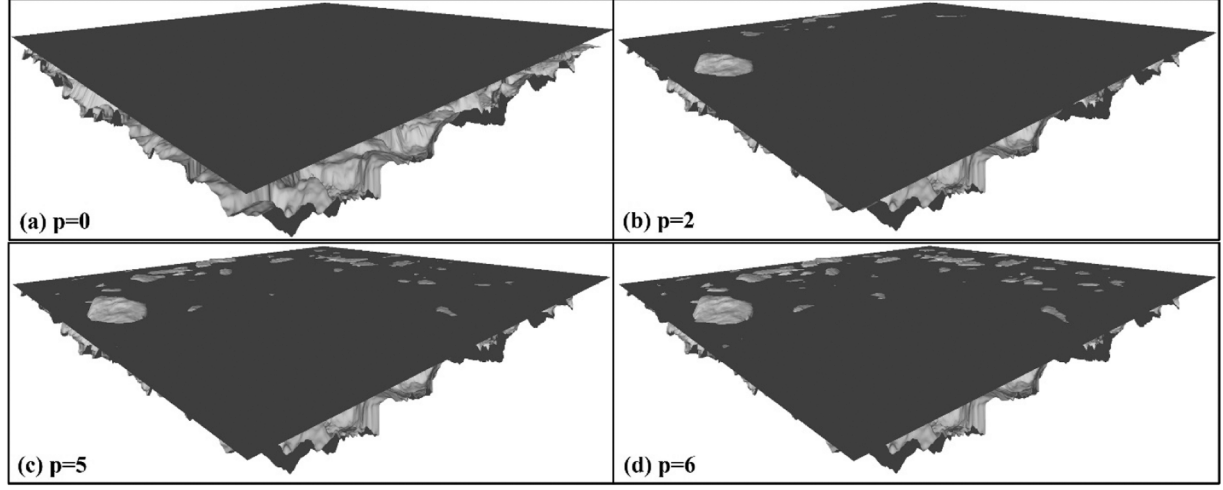


Fig. 11. Reference surface selection of the pavement.

Regarding different selections of p values for calculating MTD_p , the MAPE statistical index (Eq. (7)) is used to measure the average relative error percentage between the predicted depth values and the measured values.

$$MAPE = \frac{1}{N} \sum_{i=1}^N \frac{|MTD_e - MTD_p|}{MTD_e} \cdot 100\% \quad (7)$$

where N is the sample size. Fig. 12 shows the results of error for different

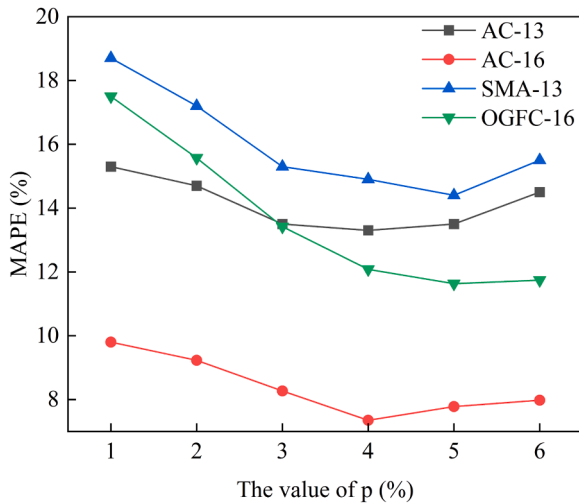


Fig. 12. The effect of p -value selection under different pavement materials.

values of p . For rough pavements such as SMA-13 with exposed stones and OGFC-16 with large voids between aggregates, a larger p value should be selected to exclude more extreme elevation points. For AC pavements with relatively fine textures and smoother surfaces, a smaller p value should be selected to more accurately predict MTD_p . Finally, the plane where the 5th percentile depth quantile value is located is selected as the texture reference plane; at this point, the prediction errors for each type of pavement are close to the minimum.

3.3.2.3. Profile-based metric. The mean profile depth (MPD) (Eq. (8)) is another widely used characterization index. The MPD divides the length of the pavement texture curve into two segments and calculates the difference between the average elevation value (h) and the average of the two peak elevation values (h_1 and h_2). Following this idea, the predicted value of the texture depth of the reconstructed pavement can be characterized by Eq. (9).

$$MPD = \frac{h_1 + h_2}{2} - h \quad (8)$$

$$MPD_p = \frac{1}{N} \sum_{i=1}^N MSD = \frac{1}{N} \sum_{i=1}^N (Z_{Al_i} - Z_{Pl_i}) \quad (9)$$

where the mean section depth (MSD) is the depth value of the average elevation of a row in the depth map (Z_{Al_i}) minus the depth value of the peak elevation (Z_{Pl_i}). N is the total number of rows in the depth map. Fig. 13 shows a schematic diagram of the calculation process of MPD_p . There are protruding aggregates above the average elevation plane. These angled particle aggregates come into contact with tires and generate friction force, thereby serving as an important impact on

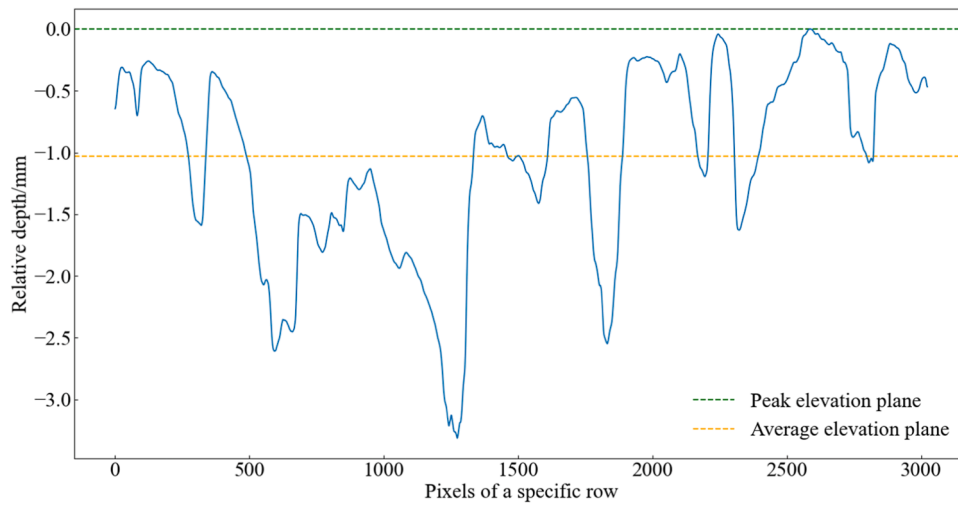


Fig. 13. Calculation diagram of MPD_p .

pavement anti-skid performance [66].

4. Results and discussion

4.1. Comparison between different training strategies

The model's reconstruction precision was compared on the RGB validation set with a resolution of 3024×3024 and 7 views for different training strategies. Fig. 14 shows the precision of the model's reconstructed point cloud after each training epoch, with and without transfer learning. The training was conducted for a total of 16 epochs, and each epoch took approximately 1.5 h to complete on the GPU. Regardless of whether transfer learning was used, the IoU increased rapidly during the initial training period. When training was conducted directly on the pavement texture RGB-D dataset without transfer learning, the peak performance of the model was achieved at approximately the 7th epoch with an IoU of 0.51. However, when transfer learning was used, the model's performance gradually increased after rapidly increasing during the early training phase. The IoU reached 0.77 at approximately the 14th epoch, which was slower than the training convergence speed without transfer learning. This is because the pretrained model's features may not be completely applicable to the target task, as the data distributions and feature representations of the two tasks may differ. Therefore, the existing features must be adjusted for a longer time to

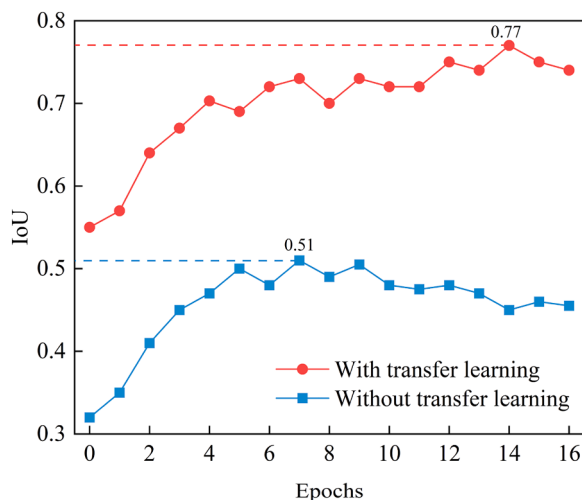


Fig. 14. Model precision after each epoch under different training strategies.

meet the requirements of the target task. Moreover, transfer learning usually requires a smaller dataset for the target task, which may be more limited than the source task. As a result, the model requires more iterations to find the optimal solution in the target task training process, leading to a slower convergence speed. Additionally, the model precision of the strategy without transfer learning decreased in the later stages of training. This is because the pavement texture RGB-D dataset has a small amount of data, and multiple rounds of training may cause the model to overfit the training set, resulting in poor performance on the validation set. After training by transfer learning for 14 epochs, the model's performance also exhibited a decreasing trend. To avoid possible overfitting, the best model was selected at the 14th epoch, where the model achieved the highest IoU of 0.77 for point cloud reconstruction precision. This model was used in the subsequent experiments.

Table 4 further compares the point cloud reconstruction precision of the model utilizing the DTU dataset for pretraining, trained directly on the RGB-D dataset, and trained by transfer learning. The pretrained model performs well because it was trained on a large-scale general dataset, which results in strong feature extraction and generalization capabilities, enabling it to perform well when handling new tasks or specific domain data. The model trained directly on the RGB-D dataset performs poorly because the dataset is small, making it difficult for the model to learn and generalize to other road scenes. In addition, the dataset may exhibit issues such as noise, sample imbalance, or missing depth values, which can negatively impact the model's performance. The model trained by transfer learning achieves the best performance by combining the powerful feature representation capabilities of the pretrained model and the specific information of the RGB-D dataset. Fine-tuning the pretrained model with the RGB-D dataset allows the model to be adapted to the data of the specific task, improving its performance in this field. Fig. 15 depicts the depth map reconstruction results of the model under different training strategies. The model trained with transfer learning can more clearly reconstruct the pavement texture contour and details and can more accurately capture the key features and shapes than the other models. This model is considered capable of effectively handling multiview image inputs, achieving high-quality depth map reconstruction results, and accurately estimating scene

Table 4

Model precision under different training strategies.

Strategies	Pre-trained model	Without transfer learning	With transfer learning
IoU	0.65	0.51	0.77

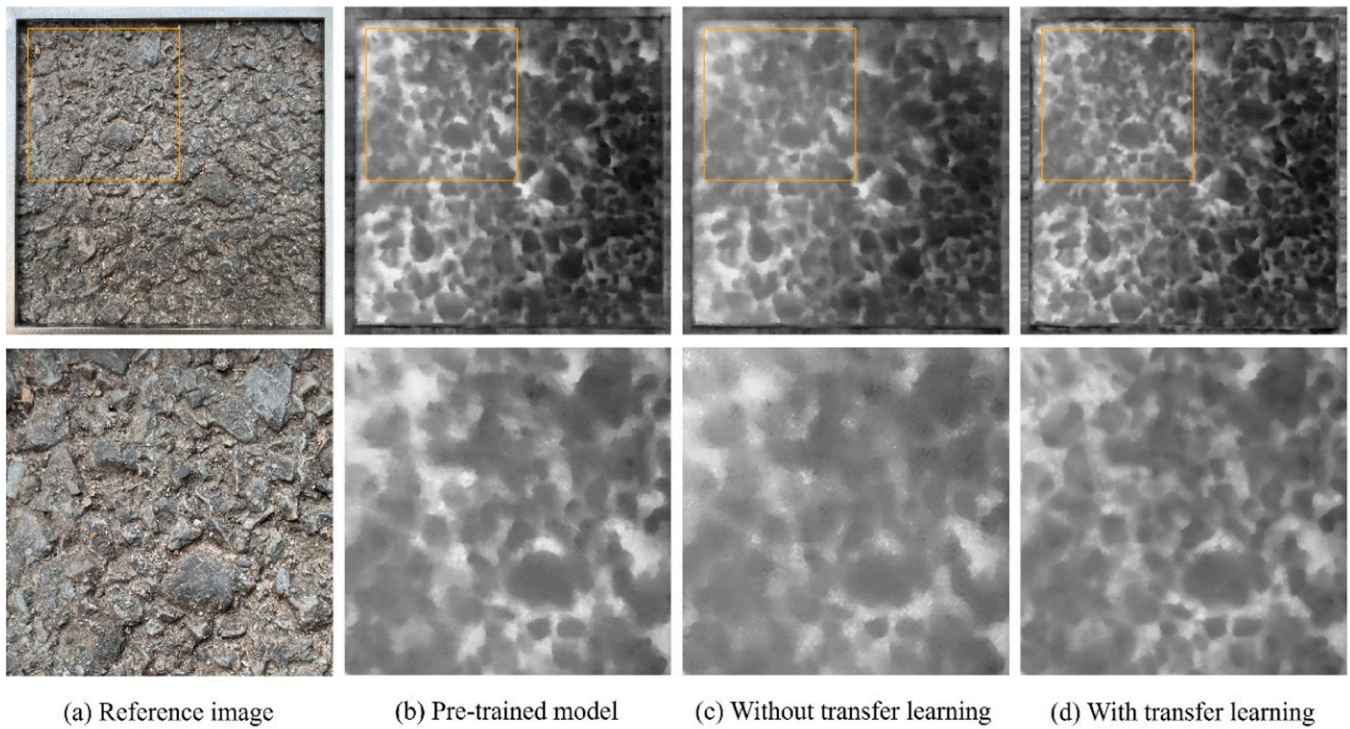


Fig. 15. Depth map reconstruction results under different training strategies.

distance.

4.2. Factors influencing model accuracy

4.2.1. View numbers and image resolution

Table 5 quantitatively compares the overlap between the reconstructed point cloud and the ground truth point cloud using Eq. (1), and Fig. 16 shows the qualitative comparison of the reconstructed depth map. As the image resolution increased from 2048×2048 to 3024×3024 , the model reconstruction quality exhibited an upward trend because a higher image resolution can provide more accurate texture details and enable fine textures to be more accurately reconstructed. However, the reconstruction quality slightly decreases at the higher resolution of 3400×3400 . This is likely because the higher-resolution reference view requires more source views to provide texture information for matching and avoid impacting the completeness of the model reconstruction. Similarly, when the number of views increased from 5 to 7, the model reconstruction quality showed an upward trend because the increase in the number of views alleviated occlusion problems and provided more texture information, thereby making the reconstruction more complete. However, when the number of views increased to 10, the model reconstruction quality decreased somewhat, which is likely due to potential errors in feature matching between the additional views, which impact the model reconstruction accuracy. In summary, the experiments showed that the optimal number of views and image resolution for model reconstruction were 7 and

3024×3024 , respectively.

4.2.2. Pavement materials

Fig. 17 shows the precision of the reconstructed point cloud under different pavement materials. The boxplot shows that the model's reconstruction precision varies for different types of pavements: the model's reconstruction performance on AC-13 pavements with smaller particle size is worse than its performance on AC-16 pavements with larger particles because the model cannot reconstruct fine textures well enough due to the potential accuracy limitations. However, the precision of the model is still within the expected range. The model's reconstruction performance on SMA pavements is better because SMA pavements have larger particle size aggregates that have more diverse shapes and textures than continuous graded aggregates of AC pavements, resulting in higher reconstruction precision. The reconstruction error on OGFC pavements arises from the complex texture structure and large number of existing holes, which make depth prediction difficult. The model tends to extract texture features of coarse aggregates that occupy a larger proportion of the pavement; as a result, some texture details are ignored [36], decreasing the reconstruction precision. Fig. 18 shows the depth map reconstruction results for different types of pavements, where pixels closer to black in color are closer to the camera imaging plane. Qualitative analysis shows that the model can accurately reconstruct the areas that protrude or are sunken relative to the road surface for pavements with a high proportion of coarse aggregates, but it ignores some finer texture details. In contrast, the reconstructed depth map provides more comprehensive detail information for pavements with smoother textures. Overall, the model's reconstruction performance is stable on various pavement materials, indicating the materials have a relatively small impact on the model's reconstruction precision. The method exhibits a satisfactory level of precision under real-world conditions.

4.3. Model performance comparison

Table 6 shows the point cloud reconstruction results of different 3D

Table 5

Reconstruction quality under different view numbers and resolutions.

Image resolution	View numbers		
	5	7	10
2048×2048	0.52	0.54	0.53
2560×2560	0.69	0.72	0.70
3024×3024	0.74	0.77	0.75
3400×3400	0.73	0.76	0.74

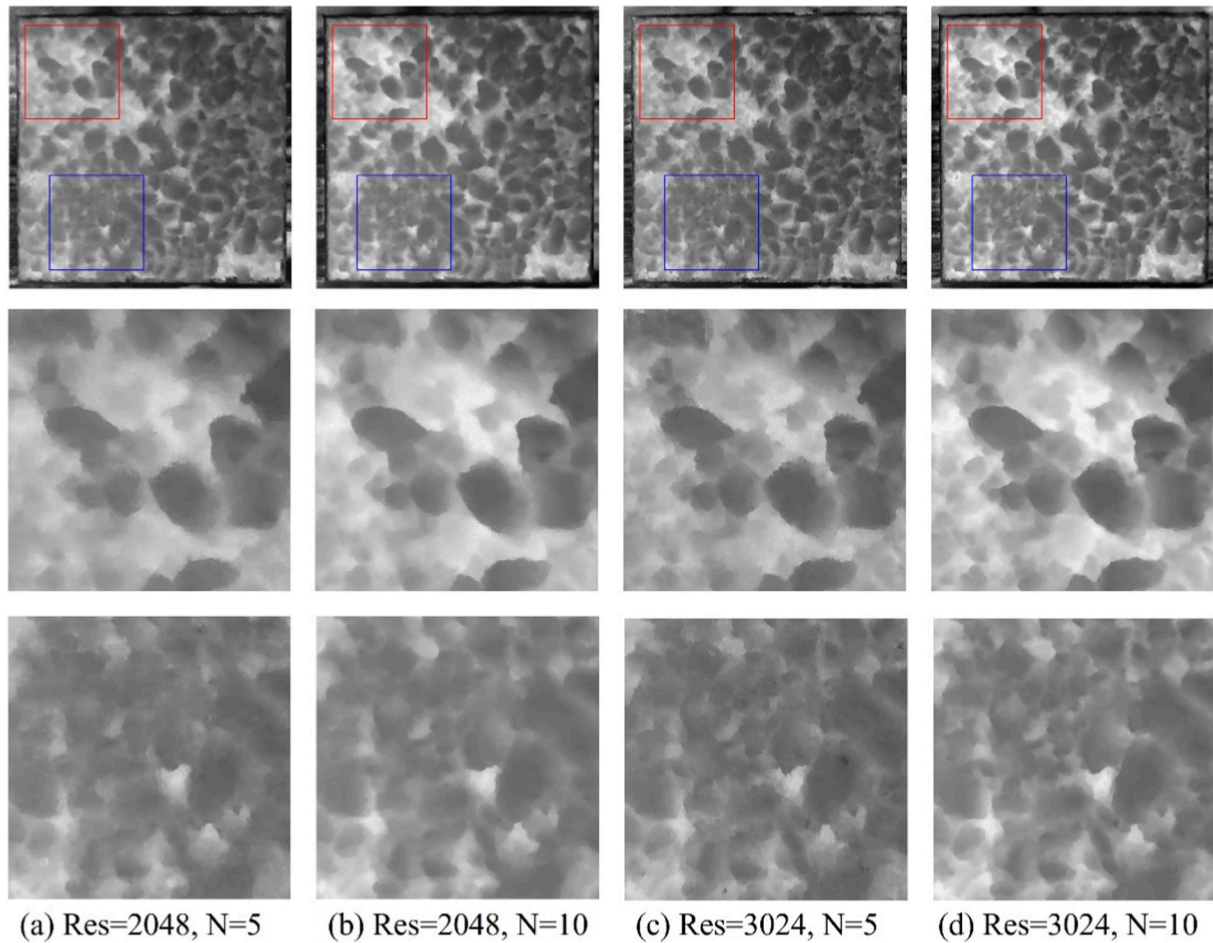


Fig. 16. Reconstruction results on the RGB validation dataset with different parameters. Top: whole depth map. Middle and bottom: zoomed local region of rectangle. N is the numbers of view. Res is the image resolution. As N increases, the reconstruction becomes more complete. As Res increases, the reconstruction becomes more accurate.

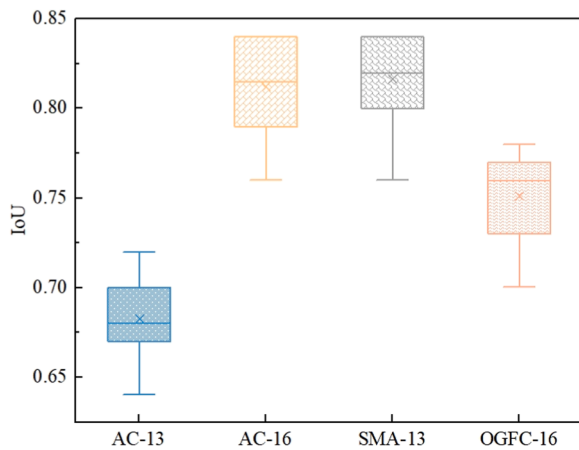


Fig. 17. Reconstruction precision on different pavement materials.

reconstruction models for pavement textures. MVSNet [47] performs poorly, not only because the simple plane-sweeping algorithm cannot effectively handle detailed textures but also because the generated depth map resolution is lower than the input image resolution, significantly reducing the precision of the reconstructed point cloud. JDACS-MS [60] uses only unsupervised-generated target signals for training. The self-generated depth maps usually have errors, which impacts the

model's learning and results in unsatisfactory reconstruction precision for detection needs. Fast-MVSNet [61] constructs a sparse cost volume to predict high-resolution sparse depth maps, resulting in depth estimation errors. These errors subsequently impact the generation and optimization of dense depth maps, leading to a decrease in reconstruction precision. COLMAP [62,63] obtains more accurate depth estimation values, but it still exhibits some problems in depth map completeness and depth continuity, and its depth estimation efficiency is lower than that of deep learning methods. PatchmatchNet, a supervised deep learning model that differs from the MVSNet pipeline network design, reduces the use of 3D cost volume regularization and integrates the Patchmatch algorithm [67] based on the assumption of dense parallel depth layers in the plane-sweeping algorithm [49] for random initialization and iterative propagation. This model achieves a good balance between reconstruction efficiency and precision. Therefore, the performance of the model selected in this paper is considered acceptable.

4.4. Effectiveness evaluation results

Partial calculation results of MTD_e , MTD_p and MPD_p are presented in Table 7, where the MTD_e value measured by the sand patch method is considered the benchmark value. The MTD_p index has good characterization ability for four different types of road materials, with an overall mean relative error of 11.72%. The absolute errors calculated based on the MTD_p index are generally less than 0.1 mm, excluding that of the OGFC pavement. These errors are close to the benchmark value, meeting the accuracy requirements for pavement texture depth detection [68].

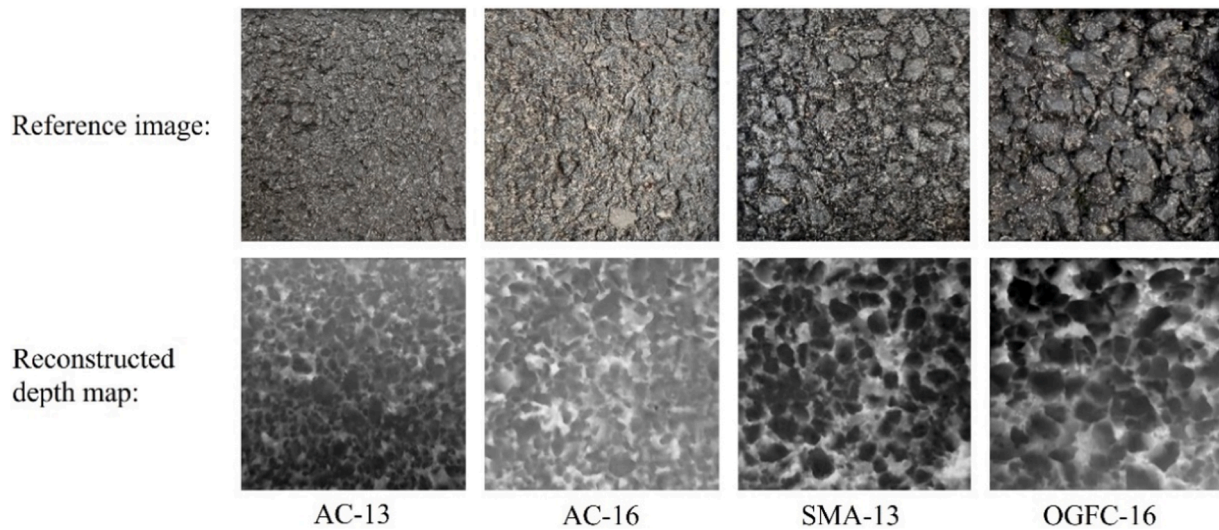


Fig. 18. Examples of reconstructed depth map for different pavement types.

Table 6

Comparison of point cloud reconstruction quality using different models.

Model	MVSNet	JDACS-MS	Fast-MVSNet	COLMAP	Ours
IoU	0.48	0.54	0.60	0.64	0.77

However, the texture depth calculated based on the MPD_p index is significantly smaller than the benchmark value, with an overall mean relative error of 18.85%. For all types of pavements, the average absolute errors of the MPD_p results are greater than 0.1 mm, and the average relative error is significantly larger than that of MTD_p . This is because the index calculated based on the two-dimensional profile is too simple to accurately characterize different textures [69]. Moreover, the calculation error of the MPD_p index is larger on SMA and OGFC pavements than on the AC pavement. This is because the aggregates in SMA and OGFC pavements are more uneven, and the profile texture fluctuations are larger. The assumed average elevation plane differs greatly from the texture at the bottom of the valley, resulting in a smaller MPD_p calculation compared to the benchmark value.

However, regardless of which index is selected, the prediction performance of the model for the OGFC pavement is worse than that for the

SMA and AC pavements, with larger absolute errors between the predicted index and the benchmark value. This is because measuring the texture depth of porous pavements using the sand patch method is not reliable; some gravel may fall into the pores, resulting in a high value of MTD_e [70]. Moreover, due to the obvious nonplanarity, large voids between aggregates, and rough surface of the OGFC pavement, the predicted depth values vary greatly between different positions in the depth map, affecting the accuracy of the model's depth value prediction when using the plane-sweeping algorithm. For some measured points on AC-13 and SMA-13 with smaller texture depths, the error between the predicted value and the benchmark value is mainly because the sand patch method is suitable for asphalt pavements with a texture depth greater than 0.25 mm [70]. The 0.3 mm clean sand cannot be fully embedded in those pavements, resulting in inaccurate MTD_e measurements.

Regardless of the selected calculation index or pavement material type, the prediction error is also impacted by the erroneous estimation of the depth information of the calibration points. This error increases to the depth values of other points due to the scale recovery process, decreasing the overall prediction accuracy. Therefore, using more calibration points and accurate calibration methods is important. The error also comes from the difference between the plane model fitted by the

Table 7

Partial calculation results of three indicators.

Materials	MTD_e (mm)	MTD_p (mm)	AE (mm)	MAE (mm)	RE (%)	MRE (%)	MPD_p (mm)	AE (mm)	MAE (mm)	RE (%)	MRE (%)
AC-13	0.57	0.65	0.08	0.086	14.04	13.48	0.46	0.11	0.118	19.30	18.57
	0.59	0.50	0.09		15.25		0.45	0.14		23.73	
	0.62	0.54	0.08		12.90		0.53	0.09		14.52	
	0.69	0.62	0.07		10.14		0.56	0.13		18.84	
	0.73	0.84	0.11		15.07		0.61	0.12		16.44	
AC-16	0.90	0.83	0.07	0.072	7.78	7.35	0.77	0.13	0.126	14.44	12.88
	0.93	0.99	0.06		6.45		0.82	0.11		11.83	
	0.96	1.04	0.08		8.33		0.84	0.12		12.50	
	1.02	1.08	0.06		5.88		0.90	0.12		11.76	
	1.08	0.99	0.09		8.33		0.93	0.15		13.89	
SMA-13	0.63	0.72	0.09	0.100	14.28	14.41	0.48	0.15	0.172	23.81	24.73
	0.67	0.56	0.11		16.42		0.49	0.18		26.87	
	0.69	0.77	0.08		11.59		0.55	0.14		20.29	
	0.73	0.61	0.12		16.44		0.54	0.19		26.03	
	0.75	0.65	0.10		13.33		0.55	0.20		26.67	
OGFC-16	2.16	1.96	0.20	0.274	9.26	11.63	1.79	0.37	0.452	17.13	19.22
	2.26	2.02	0.24		10.62		1.84	0.42		18.58	
	2.37	2.05	0.32		13.50		1.93	0.44		18.57	
	2.43	2.10	0.33		13.58		1.88	0.55		22.63	
	2.50	2.22	0.28		11.20		2.02	0.48		19.20	

algorithm and the true plane of the pavement, which leads to cumulative calculation errors.

Note : AE: Absolute error; RE: Relative error; MAE: Mean absolute error; MRE: Mean relative error.

Fig. 19 shows the fitting line graph of the data in Table 7, where R^2 represents the coefficient of determination, which is used to measure the degree of fit of the model. The value of R^2 ranges from 0 to 1. The closer it is to 1, the better the fit between the measured results and the reconstructed results. Fig. 19 illustrates that although MTD is relatively scattered, the model's predicted MTD_p and MPD_p achieved results highly correlated with the measured MTD_e . However, as discussed earlier, the error between the texture depth value calculated based on the MPD_p index and the measured benchmark value is large, and the reconstructed texture depth value based on the MPD_p calculation is not very reliable. Therefore, although a linear correlation exists between MPD_p and MTD_e , the MTD_p index is recommended for calculating the reconstructed texture depth to obtain results closer to actual measurements.

To further validate the reliability of the method, 30 measurement points were selected as samples for on-site testing on an AC-16 pavement. Fig. 20 shows the experimental results of each measurement point using the sand patch method and the proposed method. Table 8 is a comparison of the statistical data of the measurement results. Although the texture depths significantly differ between the test areas of the pavement itself, the mean values of MTD calculated using the two methods are similar. Therefore, when using the proposed method, MTD_p can be used as an approximate reference value for MTD_e .

4.5. Comparisons among different texture measurement methods

The time required to gather texture information using our method, photogrammetry, and the sand patch testing method was subjected to statistical analysis. Each method measured 10 sample points, and Table 9 records the average time and necessary measuring instruments. Following the proposed pipeline, our model takes 7 images for depth map reconstruction with a resolution of 3024×3024 . Photogrammetry is employed using PhotoScan software for point cloud reconstruction.

From the Table 9, it is evident that our method takes significantly less time to acquire asphalt pavement surface texture information compared to photogrammetry and the traditional sand patch testing method. This indicates that our proposed method efficiently performs texture measurement for asphalt pavement. Moreover, our method does not require

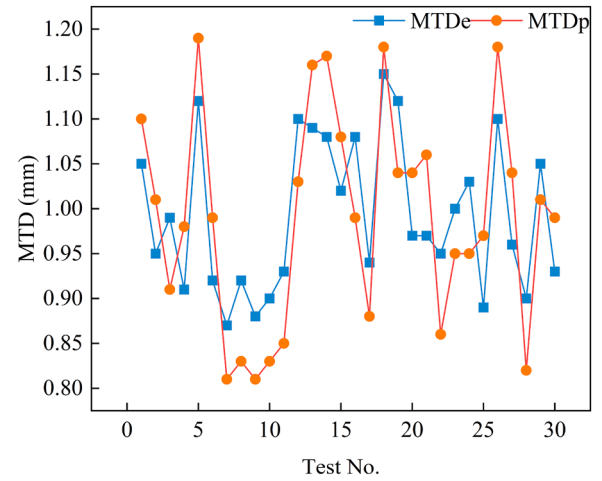


Fig. 20. MTD results of using the SPM and proposed method.

Table 8

Comparisons of the measurement and proposed methods.

	Max MTD (mm)	Min MTD (mm)	Avg MTD (mm)
MTD _e	1.15	0.81	0.9923
MTD _p	1.18	0.87	0.9903
Error	0.03	0.06	0.0020

Table 9

Comparison of different measurement methods.

Methods	Testing time	Requisite equipment
Proposed method	5 min	Camera, computer
Photogrammetry	15 min	Photographic workstation, commercial software
Sand patch method	10 min	Specialized instruments

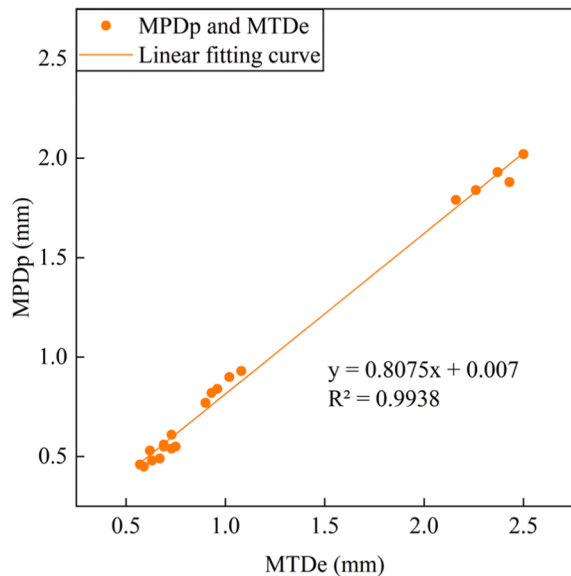
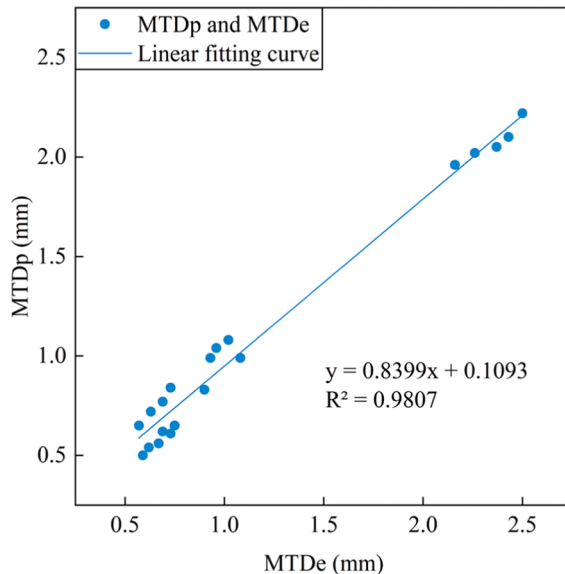


Fig. 19. Correlation between (a) MTD_e and MTD_p (b) MTD_e and MPD_p .

the additional use of a photographic workstation or commercial software for texture reconstruction, nor does it necessitate the preparation of standard sand or experimental materials. It only requires easily accessible cameras and computers for the measurements.

5. Conclusion and future works

In this study, a deep learning-based multiview stereo reconstruction method is applied for the analysis of pavement texture depth. The model performance in pavement texture reconstruction based on multiple views is discussed in detail, and a method for image processing and metric calculation of depth maps is proposed. Users can obtain information about pavement texture depth using a digital camera and a personal computer, providing a new method for measuring pavement texture depth.

The experimental results show that the well-trained deep learning network can reconstruct high-quality pavement texture models using a transfer learning strategy, with validation results indicating an improved point cloud reconstruction precision of 0.77. Additionally, the reconstruction results from various viewpoints and image resolutions suggest that the model performs best with input images at 3024×3024 resolution and using 7 multiview images. The model also exhibits consistent reconstruction precision across different pavement materials, making it suitable for pavement texture depth evaluation on various roads. Furthermore, the effectiveness evaluation results reveal that the errors in calculating texture depth using the 3D indicator MTD_p are generally within 0.1 mm, which is smaller than those obtained using the 2D indicator MPD_p . Choosing MTD_p as the calculation metric meets the standards for MTD evaluation in highway engineering testing. Finally, a comparison with two other commonly used texture measurement methods highlights the superior efficiency and convenience of the proposed approach.

Due to the difficulty of collecting RGB-D datasets and their susceptibility to environmental factors, future work will focus on improving the collection methods of training datasets. Additionally, using multiple cameras for fixed-angle shooting to derive camera poses with real-scale information and facilitate the direct reconstruction of depth maps with absolute depth values will be considered. Finally, the network must be improved to extract multiscale features from nonuniform mixtures and better predict the texture depth of different pavement types.

CRedit authorship contribution statement

Dan Han-Cheng: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Lu Bingjie:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Li Mengyu:** Writing – review & editing, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The self-created dataset in this article can provide samples upon reasonable request, but the code will not be shared due to intellectual property considerations.

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